

Distributions

Complete Guide to Probability Distributions

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Foundational Concepts

Symbol	Meaning
$f(x)$	Probability Density Function (PDF) — continuous
$P(X = x)$	Probability Mass Function (PMF) — discrete
$F(x) = P(X \leq x)$	Cumulative Distribution Function (CDF)
$S(x) = 1 - F(x)$	Survival / Reliability Function
$h(x) = \frac{f(x)}{S(x)}$	Hazard Function
$M(t) = E[e^{tX}]$	Moment Generating Function (MGF)
$\phi(t) = E[e^{itX}]$	Characteristic Function
$\mu = E[X]$	Mean

Symbol	Meaning
$\sigma^2 = \text{Var}(X) = E[(X - \mu)^2]$	Variance
$\gamma_1 = \frac{E[(X - \mu)^3]}{\sigma^3}$	Skewness
$\gamma_2 = \frac{E[(X - \mu)^4]}{\sigma^4} - 3$	Excess Kurtosis

Discrete Distributions

1. Bernoulli Distribution — $\text{Bernoulli}(p)$

A single trial with two outcomes: success (1) or failure (0).

- **PMF:** $P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}$
- **Moments:** $E[X] = p, \quad \text{Var}(X) = p(1 - p)$
- **MGF:** $M(t) = (1 - p) + pe^t$

2. Binomial Distribution — $\text{Bin}(n, p)$

Number of successes in n independent Bernoulli trials.

- **PMF:** $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n$
- **Moments:** $E[X] = np, \quad \text{Var}(X) = np(1 - p)$
- **Normal Approximation:** When $np > 5$ and $n(1 - p) > 5$: $X \approx N(np, np(1 - p))$

3. Geometric Distribution — $\text{Geo}(p)$

Number of trials until the **first** success.

- **PMF (number of trials):**

$$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, \dots$$

- **Memoryless Property:** $P(X > m + n \mid X > m) = P(X > n)$

4. Poisson Distribution — $\text{Poi}(\lambda)$

Number of events in a fixed interval given constant average rate λ .

- **PMF:** $P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$
 - **Moments:** $E[X] = \lambda, \quad \text{Var}(X) = \lambda$
 - **MGF:** $M(t) = \exp(\lambda(e^t - 1))$
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Continuous Distributions

1. Normal (Gaussian) Distribution — $N(\mu, \sigma^2)$

The cornerstone of statistics and the Central Limit Theorem.

- **PDF:**

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- **Standardized (Z):** $f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$
- **68-95-99.7 Rule:** * $P(\mu - \sigma < X < \mu + \sigma) \approx 0.6827$
 - $P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.9545$
 - $P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.9973$

2. Exponential Distribution — $\text{Exp}(\lambda)$

Time between events in a Poisson process.

- **PDF:** $f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$
- **CDF:** $F(x) = 1 - e^{-\lambda x}$
- **Moments:** $E[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}$

3. Gamma Distribution — $\text{Gamma}(\alpha, \beta)$

- **PDF:**

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0$$

- **Moments:** $E[X] = \frac{\alpha}{\beta}, \quad \text{Var}(X) = \frac{\alpha}{\beta^2}$

4. Student's t-Distribution — $t(\nu)$

Used for inference when the population variance is unknown.

- **Definition:** $T = \frac{Z}{\sqrt{V/\nu}}$ where $Z \sim N(0,1)$ and $V \sim \chi^2(\nu)$
- **Limit:** $t(\nu) \rightarrow N(0,1)$ as $\nu \rightarrow \infty$

Sampling Distributions

1. Sample Mean — $\bar{X}$

For a random sample of size n from a population with mean μ and variance σ^2 :

- **Standard Error:** $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$
- **Central Limit Theorem (CLT):** As $n \rightarrow \infty$,
$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$$

2. Sample Variance — s^2

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$:

- **Chi-Squared Relationship:**

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

Multivariate Distributions

1. Multivariate Normal — $MVN(\mu, \Sigma)$

- **PDF:**

$$f(x) = (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

- **Mahalanobis Distance:** $D^2(x) = (x-\mu)^T \Sigma^{-1}(x-\mu) \sim \chi^2(p)$

Quick Reference Table

Distribution	Support	Mean	Variance
Bernoulli(p)	$\{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	$\{0, \dots, n\}$	np	$np(1-p)$
Poisson(λ)	$\{0, 1, \dots\}$	λ	λ
Uniform(a, b)	$[a, b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal(μ, σ^2)	$\mathbb{R}$	μ	σ^2
Exponential(λ)	$\mathbb{R}^+$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma(α, β)	$\mathbb{R}^+$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
$\chi^2(\nu)$	$\mathbb{R}^+$	ν	2ν

Key Relationships

- **Normal** $\rightarrow \chi^2(\nu)$: Sum of squared standard normals.
 - **Binomial** $\xrightarrow{n \rightarrow \infty}$ **Poisson**: When p is small and $np = \lambda$.
 - **Exponential** $\rightarrow$ **Gamma**: Sum of n i.i.d. exponential variables.
 - $t(1) \equiv$ **Cauchy**: The Student's t -distribution with 1 degree of freedom.
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